ANUJ GAJJAR

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WORK EXPERIENCE

THE MITRE CORPORATION Sep 2022 – Present
Senior Embedded Software Engineer – *C++, Python, Node.js, Linux, FreeRTOS, Docker, CMake, Git* Bedford, MA

- Developed a multi-threaded, event-driven C++17 GNSS/GPS system on Linux supporting a multitude of applications
- Owned development of core components spanning signal processing pipelines, data encoding/decoding, and cryptography-related functionality
- Enhanced Python interfaces and internal tooling to improve usability, static analysis, and long-term maintainability
- Optimized CPU and memory usage, eliminating bottlenecks and meeting real-time performance constraints
- Resolved complex concurrency and timing issues, enabling stable system operation for 3+ months
- Containerized build, test, and release workflows using Docker, reducing deployment time by 3+ hours
- Authored user and developer documentation and provided technical support to cross-functional users and teams

UNIVERSITY OF NOTRE DAME May 2021 – July 2021
Advanced Wireless Research Experience (AWARE) Fellow – *C++, Python* Notre Dame, IN

- Built Python and C++ libraries for simulating, testing, and characterizing an implantable smart breast clip that provides real-time information on the condition of breast cancer tumors (presented at SPIE BiOS, Campeau et al., 2024)

NORTHEASTERN UNIVERSITY – **Optical Science Laboratory** Jul 2019 – May 2022
Undergraduate Researcher – *Python, MATLAB, PyTorch, OpenCV* Boston, MA

- Developed convolutional neural nets and optical simulations with PyTorch and MATLAB to determine the orientation of collagen molecules (published in Journal of Biomedical Optics, Alzola et al., 2021)
- Prototyped a machine learning-based image translation technique using OpenCV and PyTorch to convert confocal images of human skin, reducing microscopy acquisition time for skin cancer and disorder diagnosis

EDUCATION

NORTHEASTERN UNIVERSITY Sep 2018 – May 2022
B.S. in Electrical and Computer Engineering – GPA 3.8/4.0 Boston, MA
Activities: Sherman Center for Entrepreneurship, Generate Product Development
Coursework: Embedded Design, Operating Systems, Wireless Communication Circuits, Medical Imaging, Electromagnetics
Publications: Assessing Tissue Interrogation Volume of an Implantable Optical Sensor (SPIE BiOS, Campeau et al., 2024)
Measurement of Collagen Monomer Orientation (Journal of Biomedical Optics, Alzola et al., 2021)

PROJECTS

DISCTRACKER – *C++, FreeRTOS, Swift, Electronics & PCB design*

- Developed a functional IoT prototype for disc golf using custom electronics/PCB design and a FreeRTOS-based embedded system that streamed gameplay metrics
- Built an iOS companion application in Swift to visualize real-time data gathered over Bluetooth

MODE Radar – *C, TI RTOS, Python, Electronics & PCB design*

- Architected a millimeter-wave radar system achieving <10mm resolution for disaster environment mapping
- Designed a custom PCB and RF phased array antenna, and implemented low-level control software in C using TI RTOS
- Developed multi-process Python software for real-time data processing and communication

ShowNXT – *React Native, Node.js, GraphQL, PostgreSQL*

- Developed a full-stack recruitment platform for college athletics using React Native, GraphQL, and PostgreSQL
- Implemented core features including real-time in-app chat, user profile management, and a media-rich interface for uploading and viewing athlete highlight reels